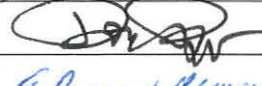
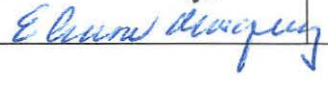


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TITLE: GXP and Non-GXP Study Report Review and Approval Form				
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☐ Vendor Study Protocol				☐ Vendor Study Report		☒ Lpath Study Report		
Lpath Study No.: LT3114-PHA-006-R			Version No.: 01		☒ Non GXP GCP		☐ GLP GMP	
Product Name: LT3114			Test Article: B3; LT3114					
Study Title: Efficacy of anti-LPA mAbs B3 and LT3114 in a rat model of acute inflammatory pain								
Study Vendor/Site: ☐ N/A LPath		Study Director: ☐ N/A Ann Gregus		Study Contributor(s): ☒ N/A				
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Executive summary of study results:				☐ See Attached		☐ N/A		
The purpose of this study was to test the efficacy of murine (B3) and humanized (LT3114) anti-lysophosphatidic acid (LPA) antibodies in reducing pain behavior in a model of acute peripheral inflammatory hyperalgesia. Male Sprague Dawley rats were injected intraplantar in the left hindpaw with carrageenan and then monitored for the presence of tactile allodynia over a period of 3 days, at which point injury typically resolves. Paw tactile sensitivity was assessed as the primary indicator of efficacy. Results showed that pretreatment with the NSAID ketorolac (30 mg/kg IP) as a study reference standard significantly attenuated carrageenan-induced tactile allodynia within the first 4 hours of testing, with reduced effects at 48h and 72h post-induction of inflammation. In contrast, there was no significant difference between groups pretreated with LT3114 or B3 as compared with control LT1023 (10 mg/kg IV) in tactile thresholds at any timepoint measured at baseline or following peripheral injury. In summary, this study demonstrates that anti-LPA mAbs LT3114 and B3 do not exhibit efficacy against a rat model of acute inflammatory pain.								

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REPORT NUMBER: LT3114-PHA-006-R	
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Efficacy of anti-LPA mAbs B3 and LT3114 in a rat model of acute inflammatory pain

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1 SUMMARY

The purpose of this study was to test the efficacy of murine (B3) and humanized (LT3114) anti-lysophosphatidic acid (LPA) antibodies in reducing pain behavior in a model of acute peripheral inflammatory hyperalgesia. Male Sprague Dawley rats were injected intraplantar in the left hindpaw with λ -carrageenan and then monitored for the presence of tactile allodynia over a period of 3 days, at which point injury typically resolves. Paw tactile sensitivity was assessed as the primary indicator of efficacy. Results showed that pretreatment with the NSAID ketorolac (30 mg/kg IP) as a study reference standard significantly attenuated carrageenan-induced tactile allodynia within the first 4 hours of testing, with reduced effects at 48h and 72h post-induction of inflammation. In contrast, there was no significant difference between groups pretreated with LT3114 or B3 as compared with control LT1023 (10 mg/kg IV) in tactile thresholds at any timepoint measured at baseline or following peripheral injury. In summary, this study demonstrates that anti-LPA mAbs LT3114 and B3 do not exhibit efficacy against a rat model of acute inflammatory pain.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is an important extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first monoclonal antibody targeting LPA (anti-LPA mAb). In preliminary studies, our anti-LPA mAb has demonstrated efficacy in attenuating pain behaviors in diabetic, chemotherapeutic, collagen-induced arthritis and sciatic nerve injury induced-neuropathic pain. Furthermore, therapy with anti-LPA Ab provides a long lasting sustained effect in accordance with the pharmacokinetic profile of the antibody. The carrageenan model of peripheral inflammation is utilized because of previously observed increases in LPA in the air pouch model. Intraplantar carrageenan produces an acute, robust and reproducible tactile allodynia. However, the role of LPA has not been described specifically in acute inflammatory pain, and this paradigm differs from partial sciatic nerve ligation injury, where the importance of LPA was initially described.

3 MATERIALS AND METHODS

3.1 Test Article and Control

Human anti-LPA antibody (LT3114) Lot# LP121402

Murine anti-LPA antibody (B3) Lot# HR111266

Human nonspecific mAb control (LT1023) Lot# LP101210

3.2 Dosing

After stable, reproducible baseline von Frey threshold measurements were obtained, five groups of rats were established (Table 1). Rats were randomized with even distribution into these treatment groups according to their baseline thresholds, yet the investigator performing behavioral testing was blinded to the dosing conditions.

Table 1 Treatment Groups/Inflammatory Pain Model

Group	No. Animals	Route	Dose	Treatment Schedule
PBS	8	IV	1 ml/kg	Day 1
IgG (LT1023) + Carrageenan	8	IV	10 mg/kg	Day 1; 1h pretreatment
LT3114 + Carrageenan	8	IV	10 mg/kg	Day 1; 1h pretreatment
B3 + Carrageenan	8	IV	10 mg/kg	Day 1; 1h pretreatment
Ketorolac + Carrageenan	8	IP	30 mg/kg	Days 1, 2, 3; 30 min pretreatment

Groups 1 and 2 were negative controls: IV PBS neutral vehicle (1 ml/kg) or a single dose of an Isotype-specific IgG for LT3114 (10 mg/kg), respectively. Groups 3 and 4 received a bolus of the anti-LPA antibodies LT3114 or B3 at 10 mg/kg IV, respectively. Group 5 received once daily Ketorolac 30 mg/kg IP as the reference standard, or positive control, for the study.

All treatment was initiated on day 1 of the study, with antibodies injected IV at a volume of 1 ml/kg as a pretreatment 1 hour before carrageenan and Ketorolac injected i.p. at 30 minutes prior to carrageenan on day 1 and 30 minutes prior to testing at 48 hours and 72 hours. Multiple injections of Ketorolac were indicated due to its shorter half-life. Tactile thresholds were measured at 1h, 2h, 3h, 4h, 24h, 48h and 72h following carrageenan. At the end of the experiment, all rats were sacrificed according to established IACUC procedures.

3.3 Tactile Response Threshold

For the assessment of tactile allodynia, Sprague-Dawley rats were placed in individual Plexiglas compartments with a wire mesh floor and allowed to acclimate to the testing apparatus for 30-60 min prior to measurement. Measurements of tactile paw withdrawal thresholds (PWT) were conducted on both left and right hindpaws by an observer blinded to the treatment conditions prior to (at baseline) and following carrageenan and/or drug delivery. Tactile PWT were measured with a series of von Frey filaments with buckling forces between 0.2 and 15 g (Stoelting, Wood Dale, IL) using the up-down method of Dixon,

as described previously (Chaplan et al 1994). Briefly, calibrated filaments beginning with a starting force of 2 g were applied to the mid-paw plantar surface until the filament was slightly bent (with a pressure corresponding to stimulation of the L4 dermatome) and maintained for up to 5 seconds. Lifting/flicking of the paw was recorded as a positive response and the next lightest filament was chosen for the next measurement. Absence of a response after 5 seconds prompted use of the next filament of increasing force. Stimuli were separated by several seconds to prevent sensitization or until the animal was stationary with paws placed flat on the grid. This method was continued until four measurements were made after an initial change in the behavior or until five consecutive negative (given the score of 15 g) or four positive (score of 0.2 g) scores occurred. The resulting sequence of positive and negative scores was used to interpolate the 50% response threshold. Rats with baseline (pre-injury) thresholds below 10 g were removed from the study.

3.4 Carrageenan Model

For the model of carrageenan-induced peripheral inflammation, rats were induced with isoflurane leading to a transient loss in the righting response and then 100 μ l of degraded λ -carrageenan (Wako Pure Chemicals Industries, Osaka, Japan) dissolved at 2% (w/v) in physiological saline was injected subcutaneously with a 30G $\frac{1}{2}$ " needle into the plantar surface (IPLT) of the left hind paw. The development of tactile allodynia was assessed at defined intervals after carrageenan injection (1h, 2h, 3h, 4h, 24h, 48h, 72h).

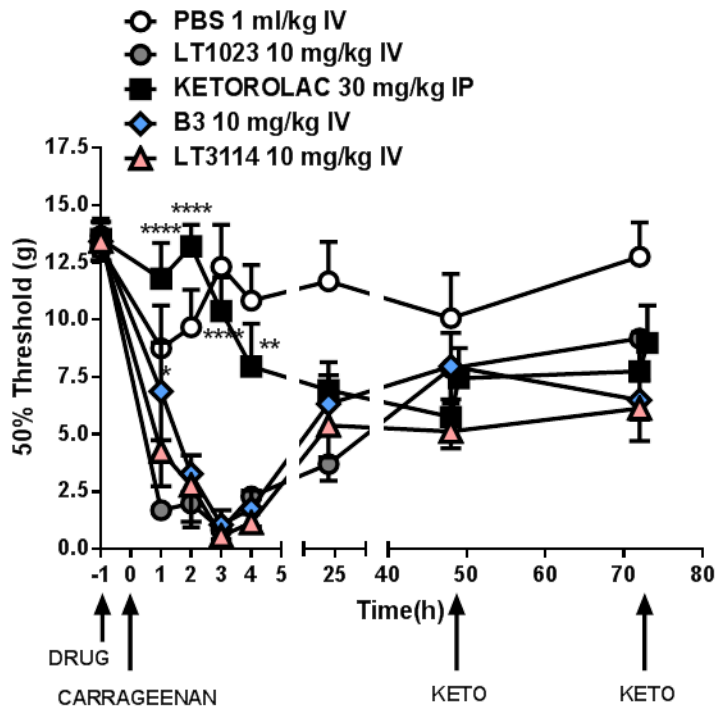
3.5 Data Analysis

All data was generated by Lpath and collected at BioTox Sciences and Lpath. GraphPad Prism software (version 6.0, Graph Software Inc, La Jolla, CA) was used to calculate mean, standard deviation (SD) and standard error of the mean as well as to graph the data. Two-way repeated measures ANOVA followed by Bonferroni correction was used for the behavioral analysis.

4 RESULTS

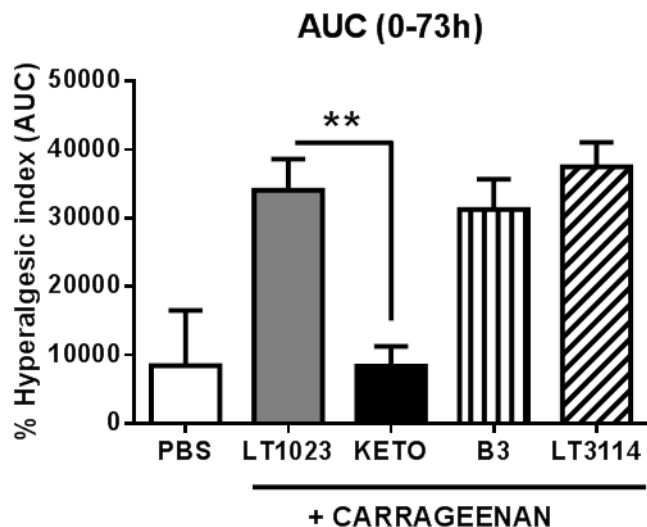
Intravenous injection with PBS vehicle produces a small, transient reduction in threshold that is not significant. Intraplantar injection with λ -carrageenan produces a robust tactile allodynia as shown by a reduction in 50% gram threshold in rats that persists for at least 72 hours (Figure 1). Pretreatment with the NSAID ketorolac (30 mg/kg IP) significantly attenuates tactile allodynia in the first 4h compared with the group receiving IgG control LT1023 (10 mg/kg IV), which is consistent with its half-life. Subsequent administrations of ketorolac at later times 48h, 72h does not markedly alter tactile thresholds. In contrast, there is no significant effect overall of either LT3114 or B3 (10 mg/kg IV) on inflammatory hyperalgesia.

Figure 1 Timecourse of carrageenan-induced tactile allodynia in rats.



LT1023, isotype control; B3, murine anti-LPA; LT3114, human anti-LPA. DRUG = IV PBS, LT1023, B3 or LT3114; KETO = Ketorolac. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.0001$ v. LT1023 by two-way repeated measures ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean $\pm$ SEM, $n = 10$ /group.

From these data, we calculated the area under the curve, or the hyperalgesic index, as graphed below (Figure 2).

Figure 2 Area under the curve (hyperalgesic index).

LT1023, isotype control; B3, murine anti-LPA; LT3114, human anti-LPA. KETO = Ketorolac. ** $p < 0.01$ v. LT1023 by two-way ANOVA with Bonferroni *post-hoc* analysis. Data are expressed as mean $\pm$ SEM, $n = 10$ /group.

5 CONCLUSION

This study demonstrates that anti-LPA mAbs LT3114 and B3 do not exhibit efficacy against a rat model of acute inflammatory pain.

6 DATA ARCHIVE

Lpath Notebook #295, pp. 6, 9-15; Binder #295, pp.16-47

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